

Synthlab Manual

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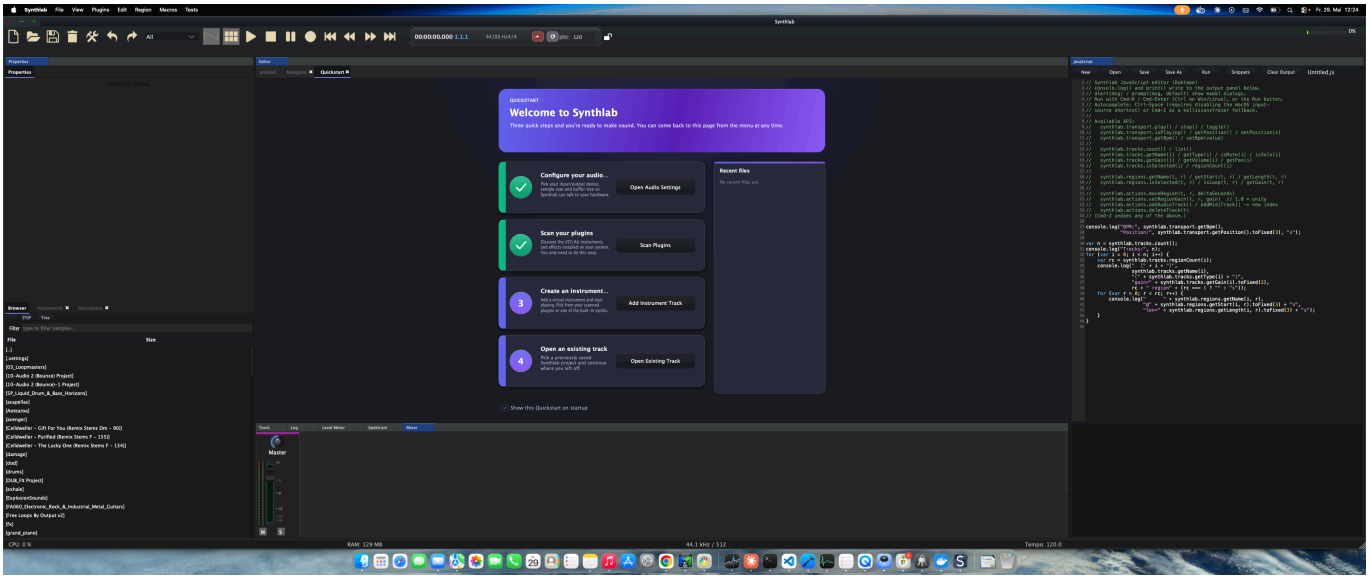
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- A. Lupo Synthesizer
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1. Getting Started

When you first launch Synthlab, you are greeted with the main application window. The interface is divided into several areas: the toolbar at the top, the Editor in the center, panels for the Browser, Properties, and Instruments on the sides, and the tool panel at the bottom which provides access to the Mixer, Level Meter, Spectrum analyzer, and Piano Roll.

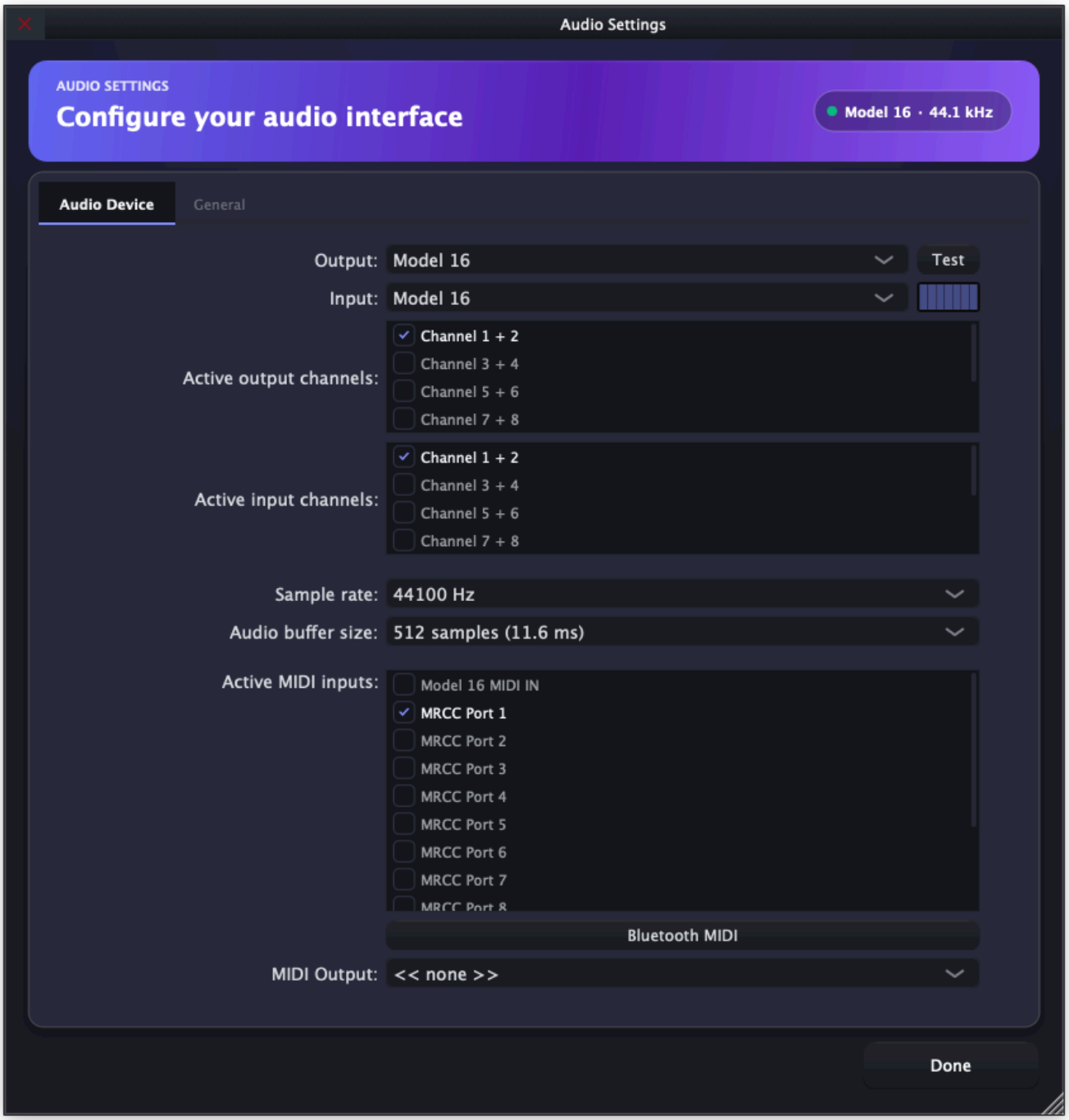


You may drag and reorder the window layout at any time, it will be saved automatically and will be restored the next time you start Synthlab.

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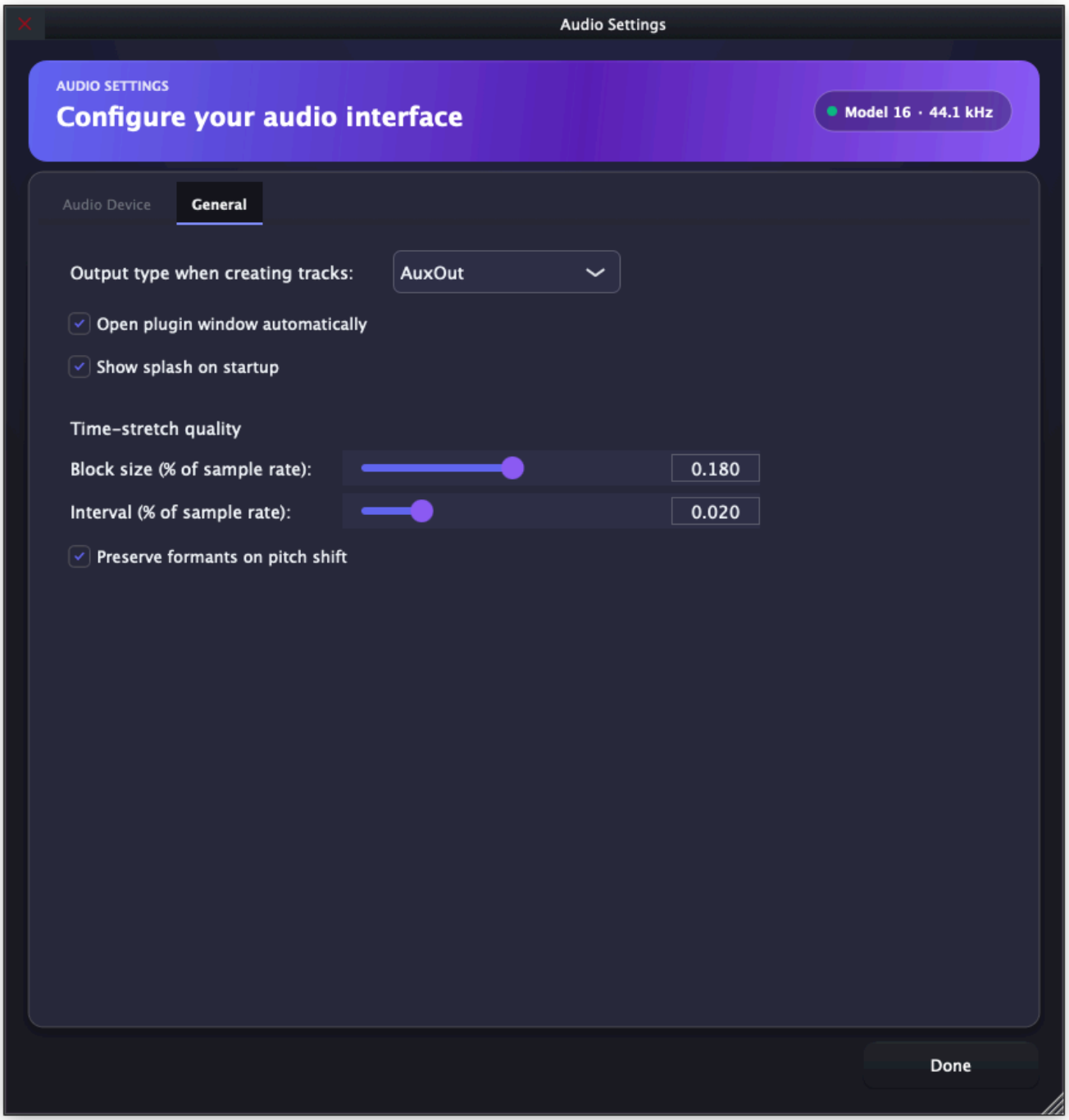
2. Settings

The Settings dialog gives you access to all global application preferences, including audio device configuration, MIDI setup, and behavior options.



3. General Settings

The General settings tab contains options that affect the overall behavior of Synthlab, such as default tempo, project path, and UI preferences.

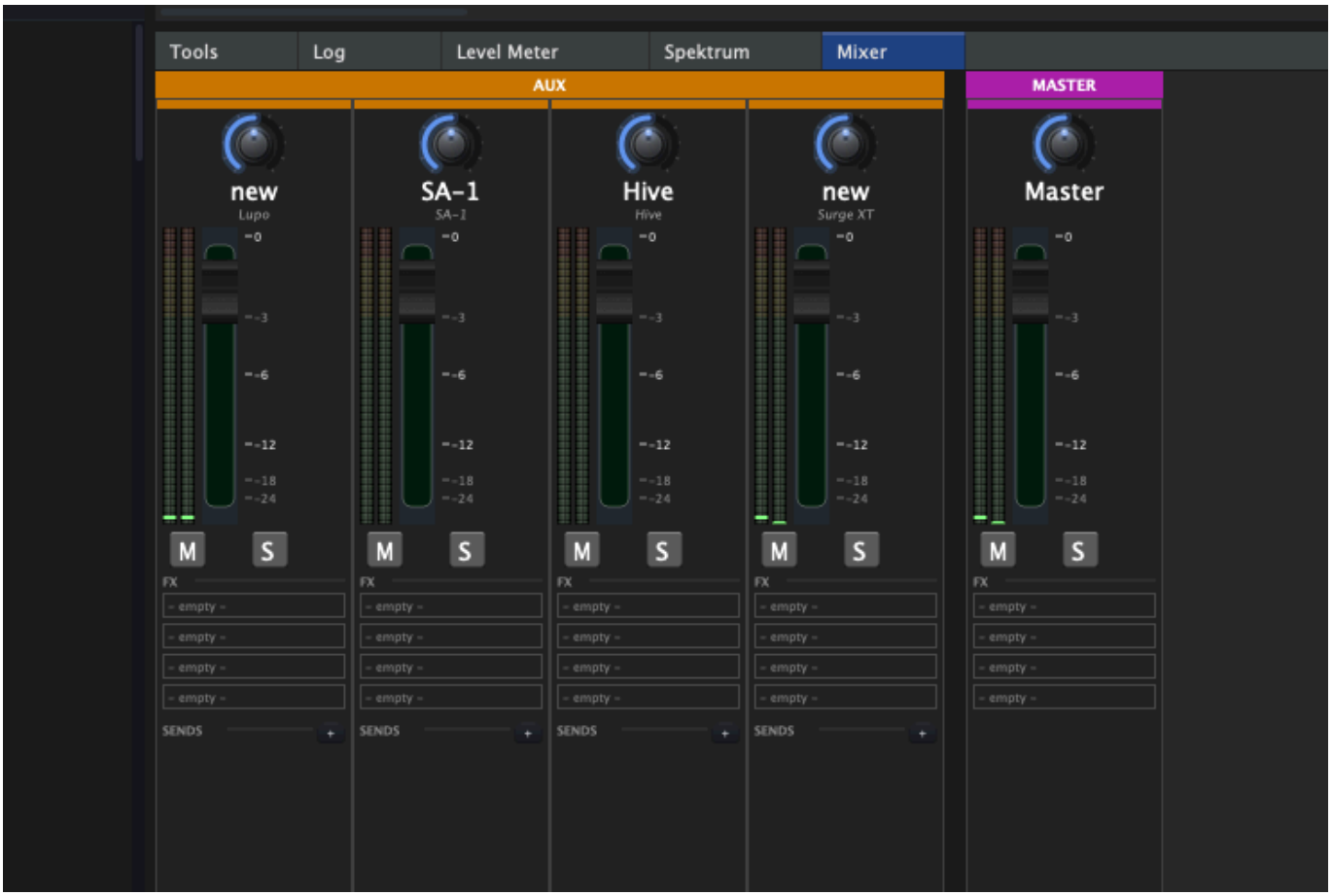


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4. Mixer

The Mixer is accessible via the bottom panel. It displays all active tracks as individual channel strips, each with a volume fader, a pan knob, Mute (**M**) and Solo (**S**) buttons, FX insert slots, and a Sends section for routing to auxiliary buses.

Channels are grouped into **AUX** outs **AUDIO** outs, **BUS** outs and a **MASTER** channel on the far right. The colored LED meters on each strip show real-time signal levels.



5. Piano Roll

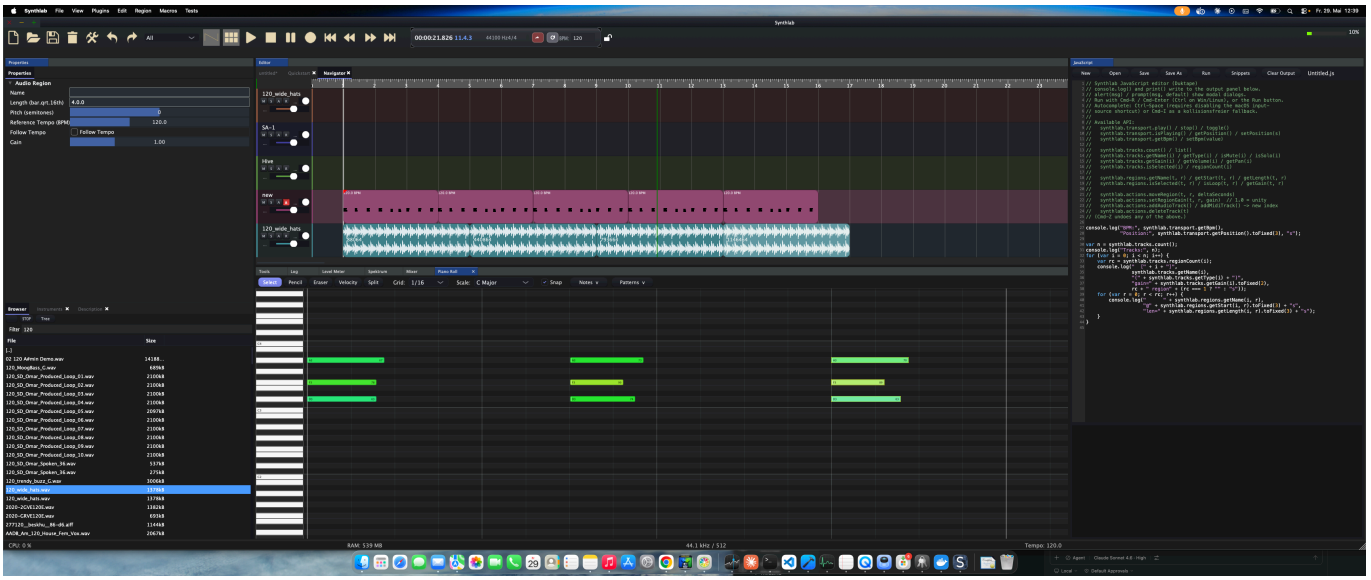
The Piano Roll editor opens in the bottom panel and allows you to draw, edit, and arrange MIDI notes. The toolbar at the top provides tools for **Select**, **Pencil**, **Eraser**, **Velocity**, and **Split**, as well as controls for the grid resolution, scale, snap, note display, and patterns.

The Piano Roll Editor provides the following shortcuts:

Shortcut	Function
CMD + A	Select all notes
Cursor Up	Move selected notes up
Cursor Down	Move selected notes down
Shift + Cursor Up	Move selected notes one octave up
Shift + Cursor Down	Move selected notes one octave down
Cursor Left	Move selected notes to the left on the raster
Cursor Right	Move selected notes to the right on the raster
Shift + Cursor Left	Shortens selected notes to the left on the raster
Shift + Cursor Right	Lengthens selected notes to the right on the raster

The keyboard on the left side of the roll indicates pitch. Notes are displayed as horizontal bars in the grid; their horizontal position represents timing and their length represents duration.

The Piano Roll shares the bottom panel with the Mixer, Level Meter, and Spectrum analyzer — click the **Piano Roll** tab to switch to it.

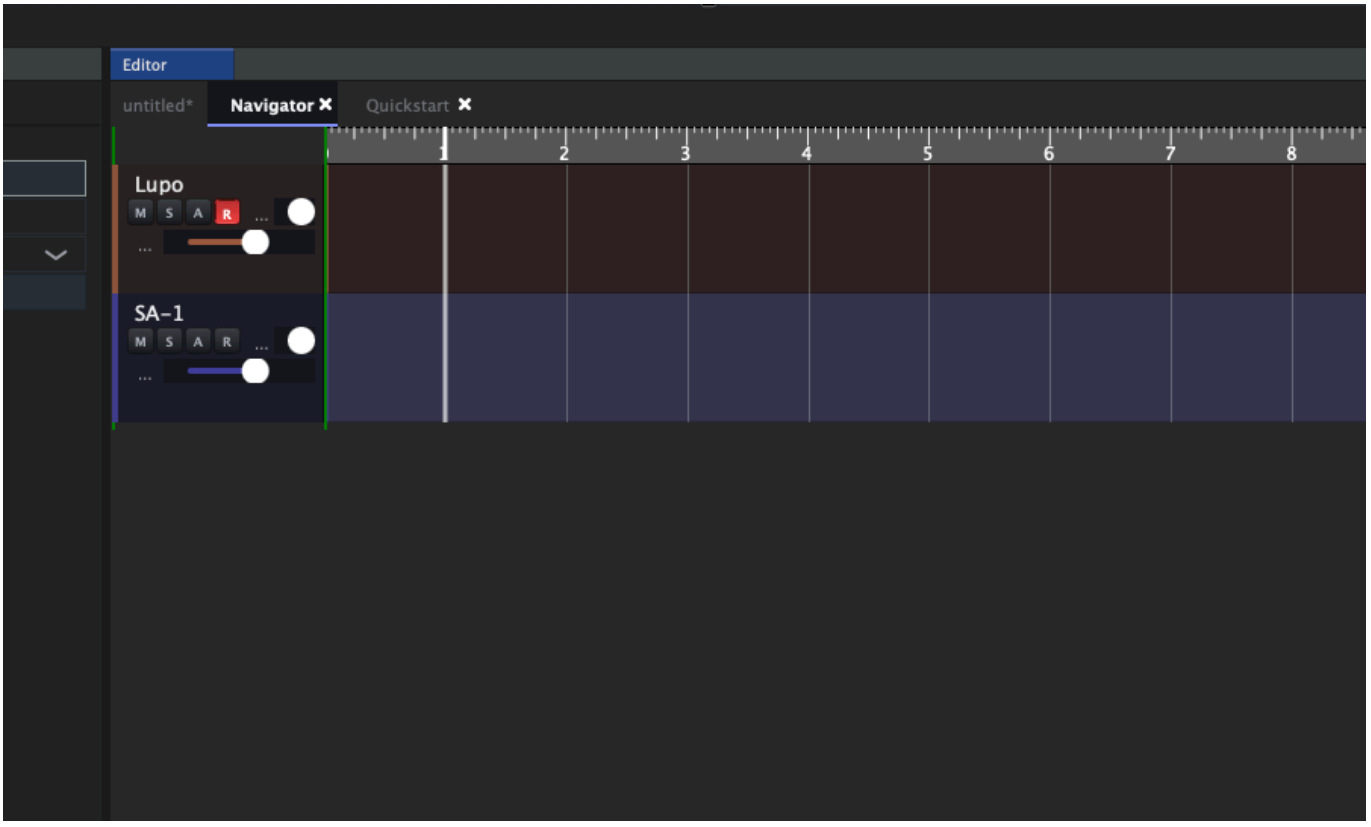


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6. Record Arming Tracks

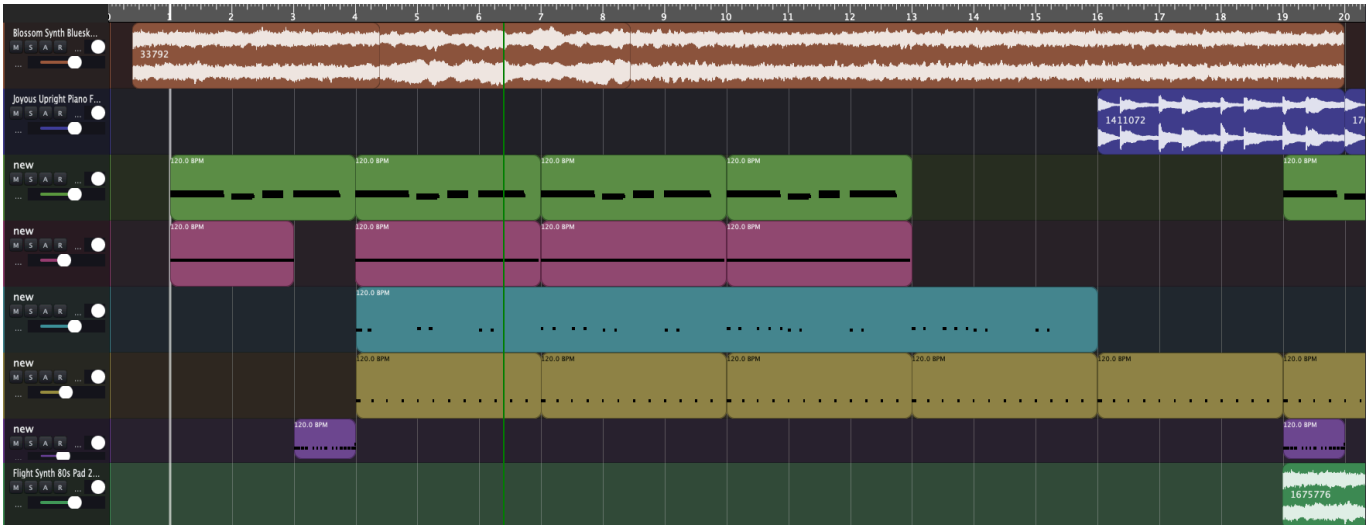
To record MIDI input onto a track, you must first arm it for recording. Click the **R** button on the track header. An armed track will show the **R** button highlighted in red.

In the example below, the **Lupo** track is armed and ready to record, while the **SA-1** track is not.



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7. Track Navigator



The Track Nabigator gives you the ability to navigate in your composition.

In order to select a track left click in the according track lane. To resize a track just grab the track separator between the tracks and drag it with the mouse. I you'd like to resize all tracks simultaneously, hold the OPTION key, while dragging.

Tracks can currently not be reordered

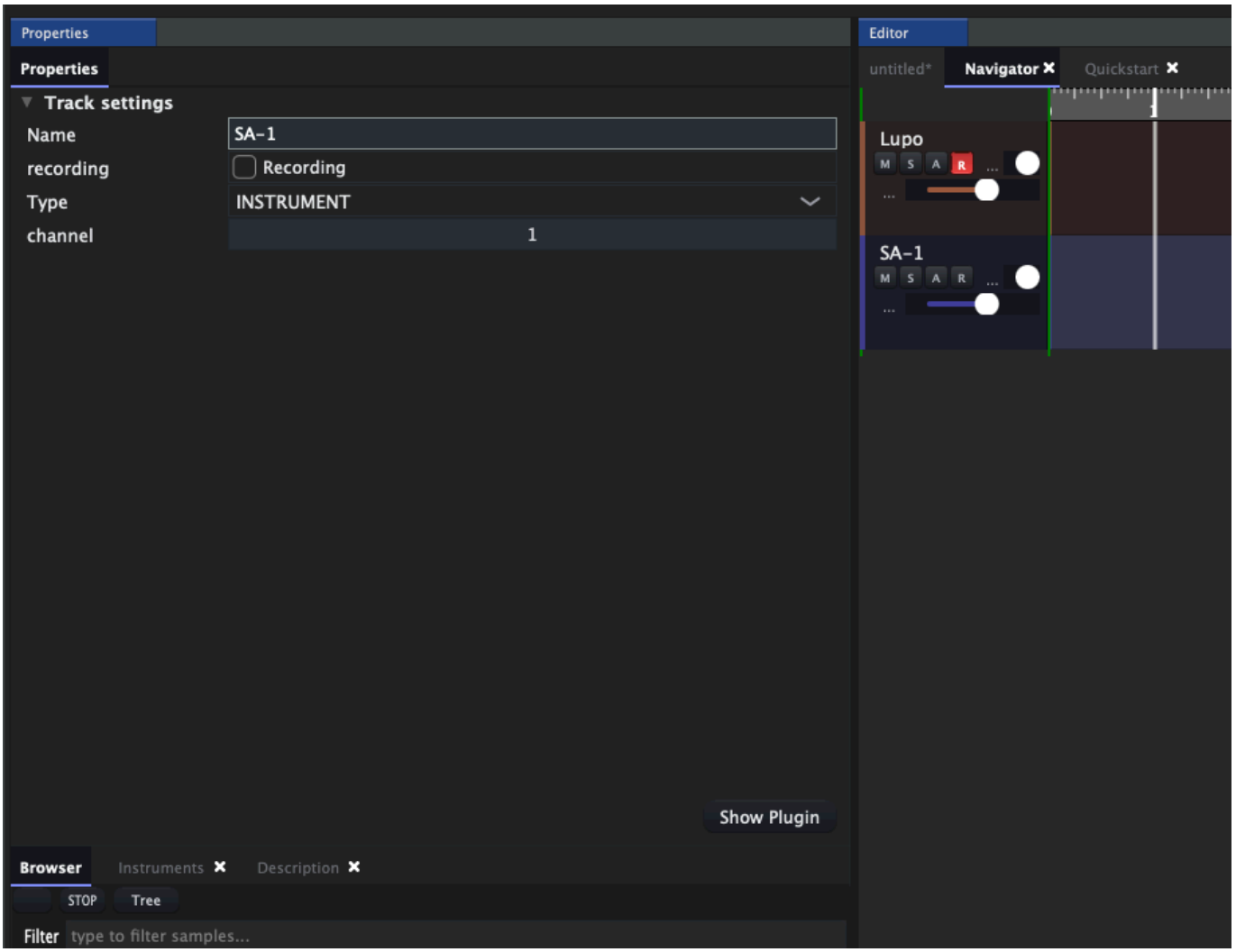
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8. Track Properties

Clicking on a trackopens the **Properties** panel on the left side of the Editor. Here you can configure:

- **Name** — the display name of the track
- **Recording** — enable or disable recording for this track
- **Type** — the track type (e.g. `INSTRUMENT`)
- **Channel** — the MIDI channel the track listens on

The **Show Plugin** button at the bottom opens the instrument plugin associated with the track.

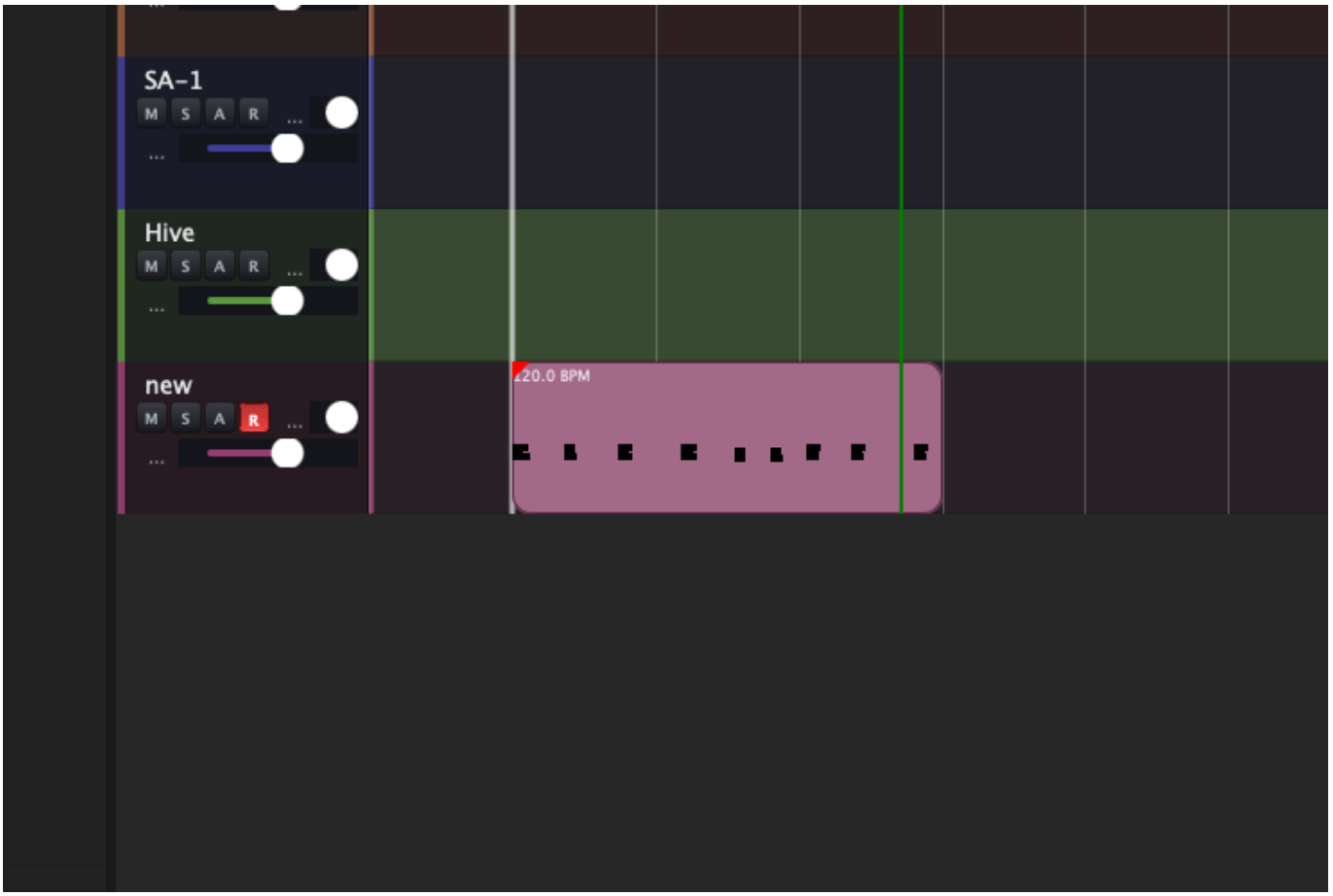


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9. Regions

A **Region** is the fundamental building block of a track in Synthlab. Each region holds MIDI note data for an instrument track (or audio data for audio tracks). Regions are displayed as colored blocks on the timeline and can be moved, resized, duplicated, or deleted.

The BPM of the project is shown in the top-left corner of each region.



Regions can be selected, moved or resized, select and move the region with the right mouse button, the right button just positions the Track marker at a new position, with the left mouse button ayou can change the selected track.

There are two types of regions.

MIDI Regions

MIDI Regions hold MIDI notes and events such as CC changes or automation data. If you double click a MIDI Region, the Piano Roll editor will be openend.

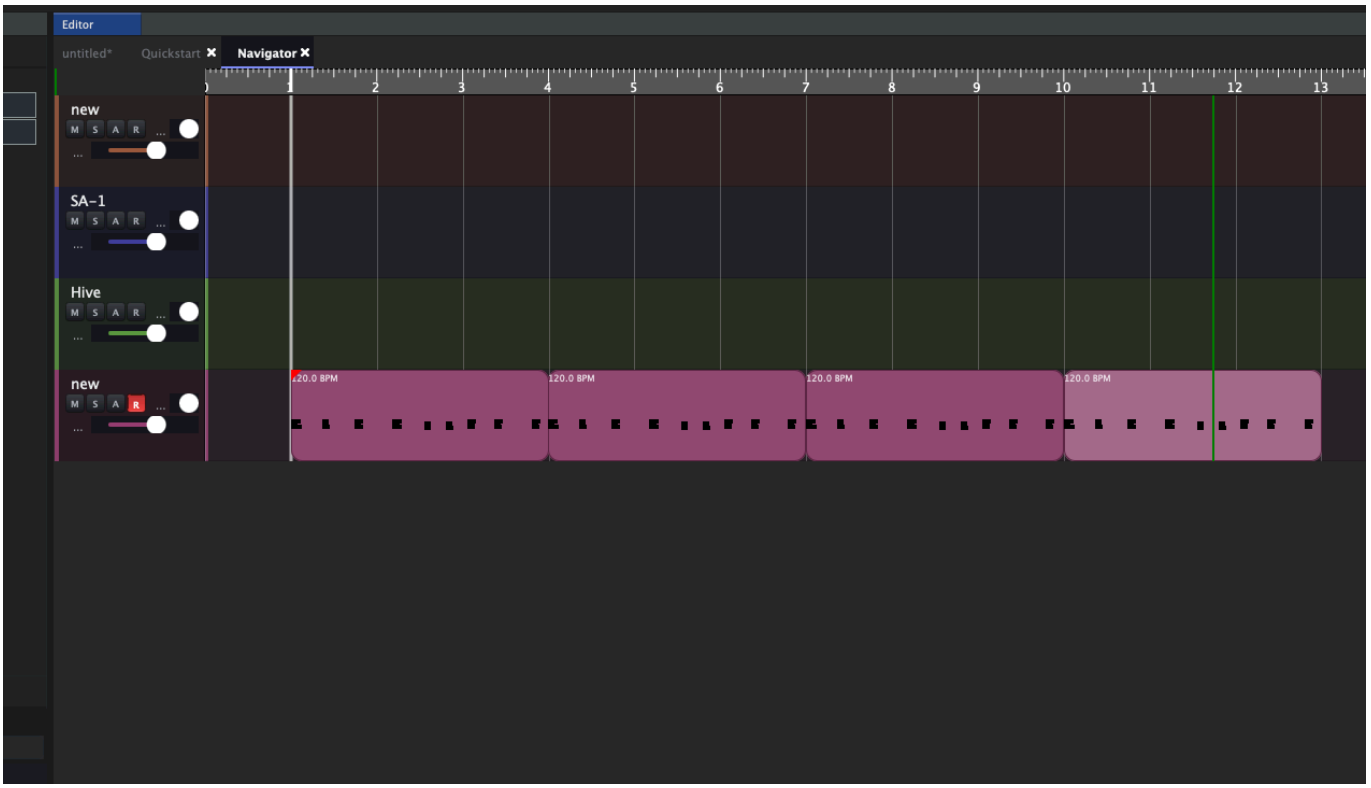
Audio Regions

Audio Regions hold Audio Data and can be moved and resized the same way MIDI regions do.

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10. Duplicating Regions

You can duplicate a MIDI region to fill a timeline quickly. Select a region and use the duplicate command to create multiple consecutive copies. The duplicated regions will appear back-to-back across the timeline, each inheriting the original's content and tempo information.



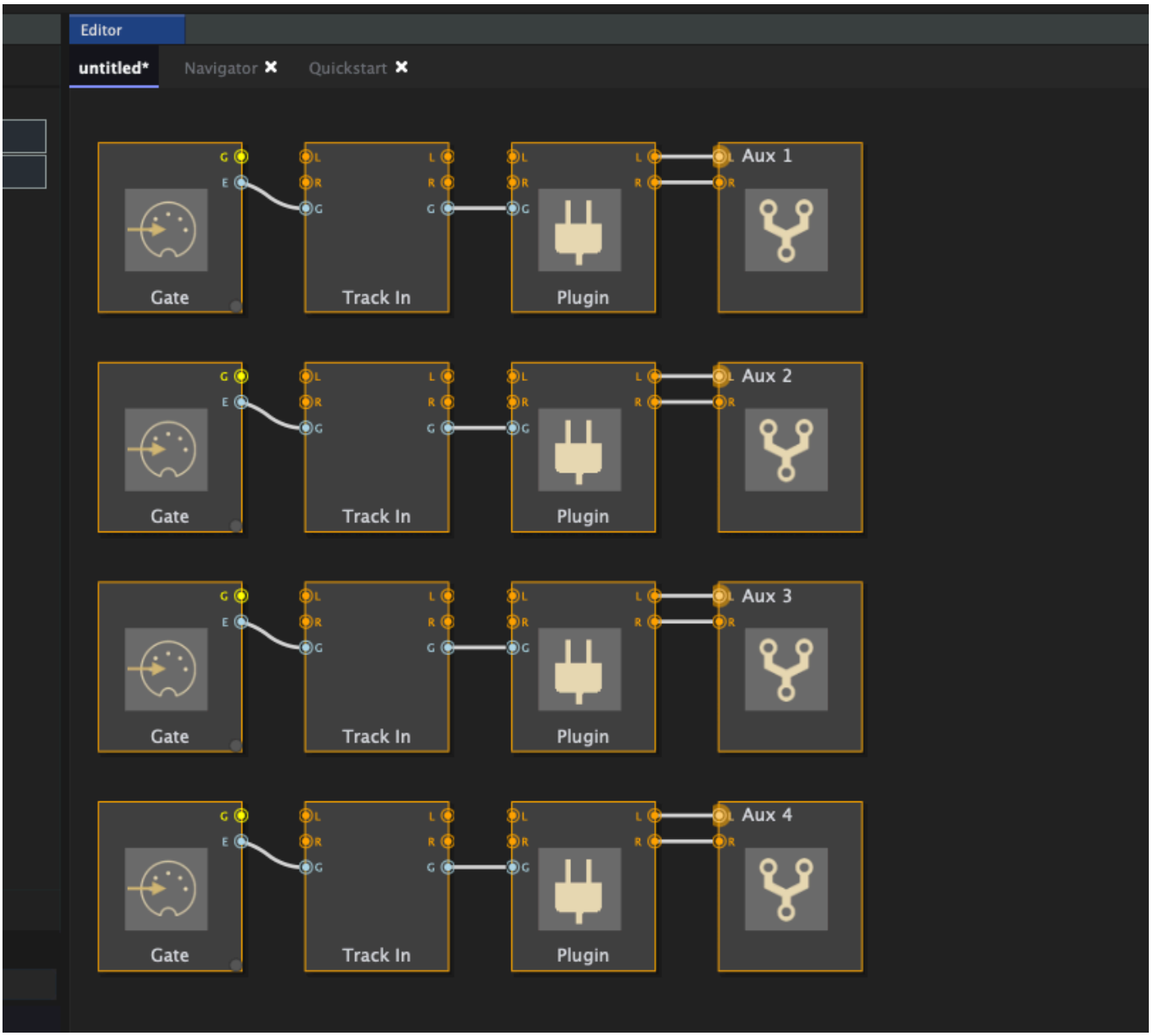
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11. Signal Structure

The **Structure** view (accessible via the Editor tabs) shows the internal signal routing of your project as a node graph. Each track row consists of:

- **Gate** — controls when the track is active
- **Track In** — receives the MIDI or audio input
- **Plugin** — the instrument or effect assigned to the track
- **Aux** — the output bus the track is routed to

This view is useful for understanding and customizing the signal flow of complex projects.

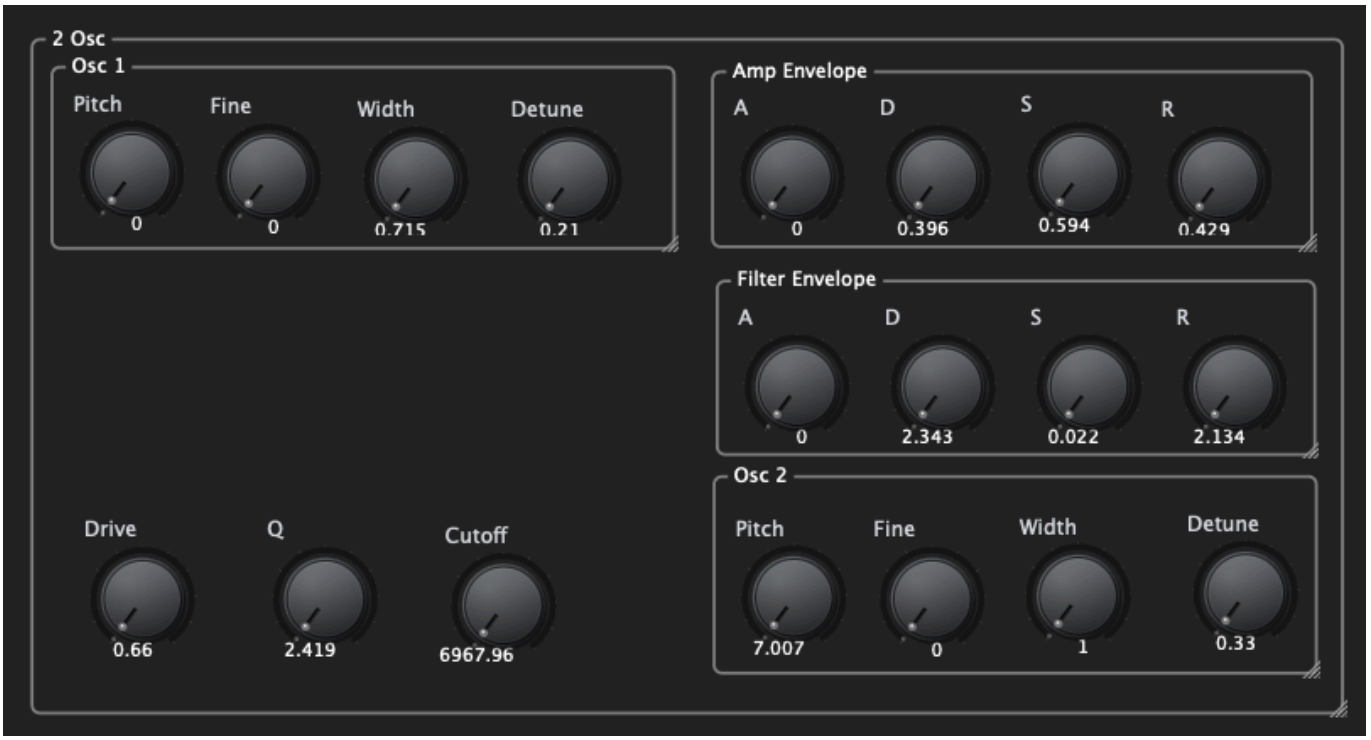


12. Building Instruments and Useful Stuff

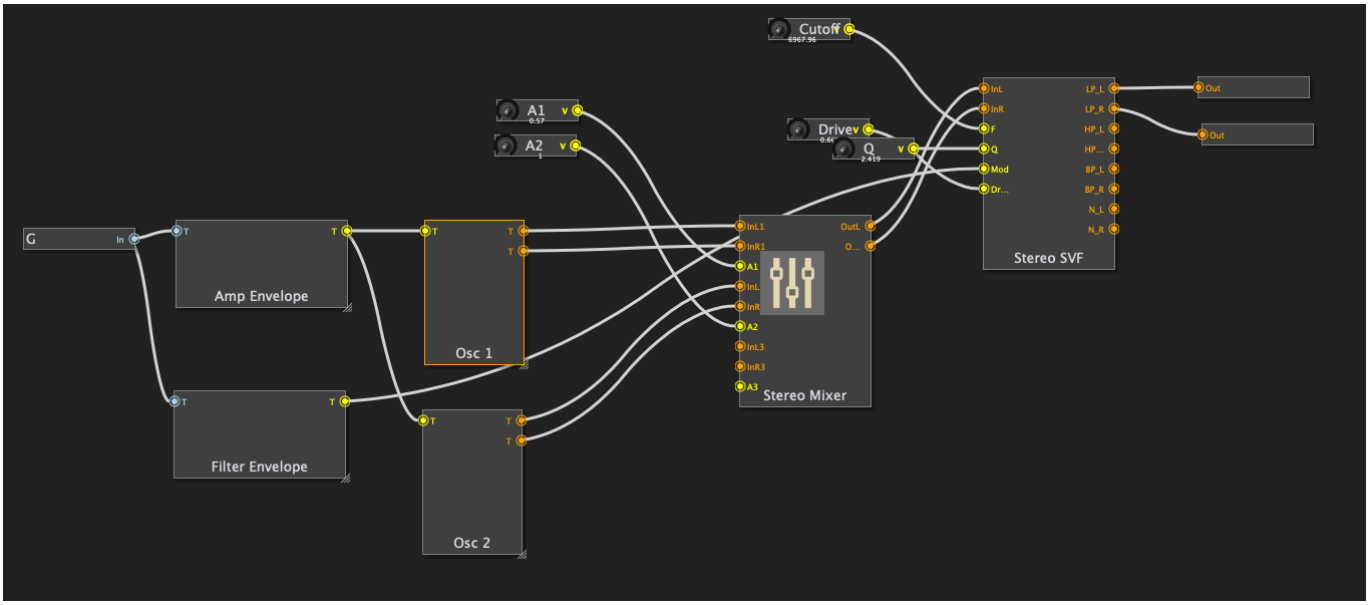
As you might have guessed, you can also build you own instruments with Synthlab in a convenient way. For this, Synthlab provides over 80 modules. We will not be discussing each module here in detail, if you want to know what each module doed se the appendix for reference.

Basics

Consider the following panel of a patch:



The patch of this Panel looks like this:



This patch contains although most of the conceptual things of Synthlab you need to know.

Appendix — Plugins and Synthesizers

A. Lupo Synthesizer

Lupo is Synthlab's built-in polyphonic synthesizer. It features four oscillators, each with controls for **Pitch**, **Fine**, **Width**, **Spread**, and **Shape** (waveform selector with options including Sawtooth, and more). A **sync** option is available on Osc 2.

The central section includes:

- **Mixer** — four channels (Ch 1–4) each with **Vol** and **Pan** knobs

- **Amplifier** — ADSR envelope (**Attack**, **Decay**, **Sustain**, **Release**) and a **Main volume** knob
- **Filter 1 & 2** — each with **Cutoff**, **Q**, **Env. Amount**, **Filter Mode** (e.g. LP), and **Character** selector. A **Cutoff link** option and **Filter mode** (Serial/Parallel) are also available.
- **Filter Env.** — a dedicated ADSR envelope for filter modulation

The right panel provides tabbed access to:

- **FX** — Reverb (Room size, Damping, Width, Dry/Wet, Freeze), Stereo Delay (Time L/R, Feedback, Mix), Portamento (Intensity, Duration), Chorus (Delay, Modulation, Feedback, Mix), and Overdrive (Drive, Mix, clipping mode)
- **EQ**
- **Sequencer**
- **Arpeggiator** — with Chord, Create, and Auto modes, plus root note and scale selectors
- **Mod Matrix**
- **Modulation**

Presets can be saved and recalled using the **Presets** dropdown at the top.



B. SA-1 Drum Sampler

SA-1 is Synthlab's drum sampler, designed for building drum kits from individual samples. It features 16 pads arranged in a 4×4 grid, each assignable to a drum hit:

Pad	Label	Pad	Label
01	KICK	02	SNARE
03	CLAP	04	RIM
05	CH	06	OH

Pad	Label	Pad	Label
07	RIDE	08	CRASH
09	TOM L	10	TOM M
11	TOM H	12	COWBL
13	SHKR	14	TAMB
15	PERC	16	FX

The waveform of the currently selected pad is displayed at the top of the window, along with metadata such as duration, sample count, sample rate, and number of channels. The MIDI note assigned to the pad is shown in the top-right corner.

Each pad can be configured individually in the right panel:

- **Volume**, **Pan**, **Pitch**, **Fine** — basic level and tuning
- **Attack**, **Release** — envelope shaping
- **Start** — sets the sample start offset
- **MASTER** — overall output volume for the sampler

Additional pad options include **Reverse** (plays the sample backwards), **Choke** (mutes other pads in the same choke group), **Load Sample** (browse for a sample file), and **Clear** (removes the assigned sample).

Kits can be saved and recalled using the **Presets** dropdown at the top.

